Highlights

Delegation cascades: exact stationary analysis of an evolutionary game with strategic chain depth

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- Delegation depth becomes a strategic variable in an evolutionary race game
- Realised harm saturates in depth while attributed harm decays geometrically
- An exact $O(Z)$ scheme replaces a chain with 3×10^{27} population states
- Two per-hand-off losses are minor alone and raise unsafe behaviour 18-fold together
- A floor under attributed liability reproduces the frontier of a depth cap

Delegation cascades: exact stationary analysis of an evolutionary game with strategic chain depth

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ABSTRACT

Software increasingly acts through chains: a principal instructs an agent, which instructs further agents, so the process that finally acts is several hand-offs from the party that wanted it. Each hand-off restates the specification imperfectly and traces responsibility back only in part. We make delegation depth a strategic variable in an evolutionary game whose interaction layer is a repeated two-player development race, and study the stationary regime of the resulting finite-population process. Two structural results are exact. Realised harm saturates in the depth while attributed harm carries one extra factor per layer, so for every attribution retention below one there is a depth beyond which each further hand-off strictly lowers the bill the principal is charged and raises the harm it causes; and any attribution rule whose share stops falling closes that shelter. The joint space of depths and specifications puts the mutation–selection chain over population states out of reach, at 3.0×10^{27} states here. We instead evaluate the interaction tensor exactly over the horizon law, reduce the process to an embedded chain on pure designs, and compute its fixation probabilities from a closed form in $O(Z)$ operations with the largest exponent factored out; the scheme is verified against a 60-digit reference, an independent implementation and the full chain. Numerically, the two per-hand-off losses raise unsafe behaviour by less than 0.02 each and to 0.32 together, and a statutory floor under attributed responsibility traces the same safety and welfare frontier as a cap on chain length.

1. Introduction

Software that acts on behalf of someone else is increasingly built as a chain. A principal issues a specification, an agent decomposes it and instructs sub-agents, and the sub-agents call further tools and agents. The process that finally takes an action is then several hand-offs away from the party that wanted it taken [1, 2]. Governance of such systems is written almost entirely at the two ends of the chain: safety training and evaluation act on the model, and liability, procurement rules and disclosure requirements act on the principal [3, 4]. The hand-offs in between are treated as plumbing.[2]

A hand-off does two things a model can represent. It transmits a specification imperfectly, because each layer restates the task in its own terms and the part most easily dropped is a qualifying clause rather than the task itself. Degradation under repeated re-encoding is a robust property of transmission chains: in human cultural transmission the distinctions that nothing selects for are the first to be lost [6], and the same decay appears in recursive machine generation [7, 8] and in chains of language models that each restate what the previous one wrote [9, 10]. It has been measured on instructions directly: across fourteen frontier models, adherence to explicitly stated instructions falls with every additional turn, from 0.88 to 0.71 over three turns for the strongest of them [11]. Failure data from deployed agent frameworks show the same two facts in the field, namely that specification errors between agents are the dominant failure class and that a linear chain of hand-offs degrades more than any other topology [12, 13]. A hand-off also attenuates responsibility, because harm that reaches the world through several intermediaries is traced back to the party at the top only in part. The gap this opens has been named for the case of a machine whose behaviour its operator

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[2] We call the result a delegation cascade. The term denotes compounding per-hand-off loss along a single chain from a principal to the party that acts, and not the informational cascade of Bikhchandani et al. [5], in which independent decision makers rationally discard their own signals after observing what their predecessors did.

cannot predict [14]; ours is a different source of the same gap, structural rather than epistemic, and one a principal can widen at will by adding a layer. Both effects are per-layer, so both compound geometrically in the number of hand-offs, and a cost that compounds that way is the classical reason a long hierarchy is expected not to survive competition [15].

The two effects push a principal in opposite directions. Imperfect transmission is a private cost, so a principal who is held responsible for what its chain does has every reason to keep the chain short. Imperfect attribution is a private benefit, since depth buys a discount on the bill. Which force wins is not a matter of taste, and neither is the policy question that follows: whether an instrument that improves transmission, one that restores attribution, or one that caps the length of the chain does more for the same effort.

1.1. Contribution

We answer both questions by making delegation depth a strategic variable in an evolutionary game. The interaction layer is the reduced four-strategy development race of Fernández Domingos and Han [16], used without modification, so that every depth effect we report is attributable to the delegation layer built on top of it. A design is a pair (d, s): a number of hand-offs and the specification the principal issues. One hand-off acts on the specification by a row-stochastic kernel, and the harm a chain causes is charged to its principal at the rate ϕ^d. Principals do not optimise; designs that pay better are copied more, in the standard finite-population dynamics of evolutionary game theory [17–19]. The contribution is an explanation together with the scheme that makes it computable, and it has three parts.

Structure. Realised harm saturates in the depth because the specification is eventually forgotten, while attributed harm carries the extra factor ϕ per layer and is bounded below by nothing. Theorem 4 turns that asymmetry into a threshold: for every $\phi < 1$ there is a depth beyond which each further hand-off strictly reduces the harm the principal is charged for while strictly increasing the harm it causes. Beyond that depth the invasion condition for one more layer stops depending on liability in the usual direction, and raising the penalty makes the deeper design more attractive rather than less (Proposition 5). What closes the shelter is not the level of attribution but whether it keeps falling (Proposition 8).

Computation. The joint design space has $n = 4(\bar{D} + 1)$ elements, which is 28 at the depth ceiling $\bar{D} = 6$ used here. The mutation–selection process over population states is then a Markov chain on $\binom{Z+n-1}{n-1}$ states, which is 3.0×10^{27} at $Z = 100$ and beyond any direct treatment. Section 5 sets out the scheme we use instead. It has three ingredients. The interaction tensor is evaluated exactly over the horizon law rather than sampled, so the payoff matrices carry no simulation noise. The process is reduced to an embedded chain on the n pure designs in the small-mutation limit. And the $n(n-1)$ fixation probabilities that chain needs are evaluated from a closed form in $O(Z)$ operations each, with the largest exponent factored out of a sum whose terms otherwise overflow double precision by three orders of magnitude in the exponent. One stationary regime costs 28 milliseconds and the whole 21×21 parameter plane of Section 6.2 costs under a minute. Table 3 verifies the scheme against a 60-digit evaluation of the same sum, against an independent implementation and against the full mutation–selection chain on a design space small enough to carry it.

Numerical results. The two mechanisms do not add. With the liability calibrated so that the depth-zero race is already safe, imperfect transmission alone yields a long-run unsafe frequency of 0.018 and imperfect attribution alone 0.012, while the two together yield 0.322; the interaction, +0.293, is 91% of the joint effect. A counterfactual that holds the population composition fixed shows why. Under full pass-through the population answers transmission loss by shortening its chains from six hand-offs to two, and that answer, rather than any change of behaviour at a given depth, is what per-layer attribution removes. In the resulting equilibrium every principal issues the safest available specification and the population executes it only 26% of the time. Among the four governance instruments we compare, restoring attribution has a threshold rather than a gradient, capping chain length is monotone and cheap, and improving transmission fidelity is neither monotone nor safe in the small. A statutory floor under attributed responsibility traces the same safety and welfare frontier as a depth cap, with a mean gap of 0.005 in unsafe frequency at matched welfare, and it asks a regulator to count no hand-offs.

1.2. Assumptions

Three assumptions belong at the outset rather than at the end. The chain transmits one specification per race, so the executed designs of the two seats are independent draws and every payoff is bilinear; a model in which layers renegotiate during play would not have this structure. The erosion kernel is exogenous and is not itself under selection, so we describe what a population does under a given transmission technology and not how that technology evolves. And attribution decays geometrically, which we treat as the definition of per-layer liability rather than as an empirical claim; Section 6.5 reports what changes when the rule is replaced, and Section 7.1 states what the model still cannot represent.

The paper is organised as follows. Section 2 places the model in the literature. Section 3 defines the interaction layer, the delegation chain and the functionals every reported number is built from, and Section 4 proves the structural results. Section 5 gives the numerical scheme, its cost and its verification, and Section 6 reports what the model does: the decomposition of the two mechanisms, the attribution–erosion plane, the four governance instruments and the robustness checks. Section 7 interprets and limits those findings and Section 8 concludes; Appendix A derives the closed form of Section 5.3.

2. Related work

Evolutionary models of development races. That a development race trades safety for speed, and that the trade worsens as the field gets more crowded, is the argument of Armstrong et al. [20]; treating that race as an evolutionary game played in a population of developers is due to Han et al. [21]. The reduced four-strategy race we use as our interaction layer is the model of Fernández Domingos and Han [16], in which participants repeatedly choose Safe or Unsafe, unsafe development pays more per round and advances faster, and a leader that cut corners carries a private risk of a setback. That line of work has since examined sanctioning against rewarding [22], heterogeneous and networked populations [23], voluntary safety commitments [24], the incentives of the auditors a regulator would rely on [25] and the trust of the users who face the result [26]. All of it places the strategic agent in the race. We leave the race exactly as it is and ask what happens when the party whose design is selected is several hand-offs removed from the party that acts.

Delegation and hierarchy. Behavioural work has shown that delegating a decision to an artificial agent changes what people choose [27] and that people who program an artificial delegate transmit their intent accurately but not precisely [28]; that literature studies a single delegation step. Accountability degrades because of the structure of an algorithmic supply chain rather than the conduct of any single actor in it [29], and the accountability problems raised specifically by multi-step agentic delegation have begun to be described [30–32], as far as we are aware without a formal model in which the depth of the chain is itself selected. That authority degrades as it passes down a hierarchy is classical in the economics of organisation, appearing as the loss of control that bounds firm size [33, 34], as the cost of decentralised information processing [35], and as the question of how many tiers to build when every tier costs control [15, 36]. Diluted responsibility in teams is equally classical [37], and experiments find that principals delegate precisely to shed blame: an unfair choice routed through an agent draws less punishment, and is made more often, than the same choice made directly [38, 39]. Closest to the mechanism we report is the judgment-proof problem [40], in which an injurer whose assets fall short of the harm it can cause takes too little care because the part of the bill it will never be presented with does not enter its calculation; that status can be engineered rather than suffered [41]. Depth is a way of becoming judgment proof by construction, and unlike a balance sheet it has no natural bound. Putting a structural property under selection alongside strategy is the subject of coevolutionary games [42], where that property is typically the interaction network, reputation or the hierarchy itself [43]; the number of intermediaries between a strategy and its execution has not, to our knowledge, been placed under selection before.

Computation of stationary regimes. The finite-population pairwise-comparison process and its fixation probabilities are standard [19, 44], as is the small-mutation limit that reduces the mutation–selection chain to an embedded chain on the pure strategies [45–47]; how small a mutation rate that limit needs has been characterised analytically for imitation processes at arbitrary selection intensity [48]. General-purpose implementations organise the computation around the full population state space [49], which is exactly what a design space of the size used here forbids. Our scheme keeps the analytical route and adds the numerical care that a strategy space of this size and a selection intensity of this strength require; Section 5 makes the comparison quantitative.

3. The model

3.1. The interaction layer

Two seats play the repeated race of Fernández Domingos and Han [16]. In each round a seat plays Safe or Unsafe. Unsafe pays more in the stage game, with round payoffs 1.0 and 0.6 for a safe player against a safe and an unsafe opponent against 2.4 and 2.0 for an unsafe one, and it contributes 1.5 rather than 1 to race progress. The race lasts

Table 1
Interaction layer at depth zero. Task payoff $a(i,j)$ and expected number of Unsafe actions $m(i,j)$ of the focal design i against j, evaluated exactly over the horizon law. Reading down a column, m never falls: the erosion order is a harm order.

	$a(i,j)$				$m(i,j)$			
i	AS	CS	CAS	AU	AS	CS	CAS	AU
AS	59.00	59.00	8.60	5.40	0.00	0.00	0.00	0.00
CS	59.00	59.00	26.64	16.60	0.00	0.00	4.22	8.00
CAS	101.77	61.63	27.20	27.20	1.00	4.78	9.00	9.00
AU	48.64	47.36	27.20	27.20	9.00	9.00	9.00	9.00

T rounds, with T guaranteed to be at least five and stopping with probability 0.2 per round thereafter, so $\mathbb{E}[T] = 9$. The leader in accumulated progress at the horizon takes a prize of 100, split on a tie, and a seat that leads or ties loses everything it has accumulated with probability $p_{\max} n^U/T$, where n^U is its number of unsafe rounds and $p_{\max} = 0.6$.

Following the source study we work with four reduced designs, written here in the order in which their specifications lose clauses: AS is unconditionally safe, CS opens Safe and then copies the opponent, CAS opens Unsafe and then copies, and AU is unconditionally unsafe. All four are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is the horizon. Every matchup is therefore evaluated by exact expectation over the horizon law rather than by sampling, which removes simulation noise from the payoff matrices entirely; the cost of the alternative is quantified in Section 5.6. Write $a(i,j)$ for the expected task payoff of design i against j, $m(i,j)$ for its expected number of Unsafe actions and $u(i,j)$ for its expected fraction of Unsafe rounds.

Reading down any column of m in Table 1, the entries never fall. This monotonicity is the structural fact on which everything below rests, and we record it as a property of the interaction layer rather than as an assumption of ours. It holds round by round and not merely in expectation, which is what makes it independent of every numerical choice in the paragraph above.

Lemma 1 (The erosion order is a harm order). *Write $a_t^{i|j} \in \{\mathrm{S},\mathrm{U}\}$ for the action taken in round t by design i against design j, and order the two actions $\mathrm{S} < \mathrm{U}$. For every opponent j, every horizon n and every round $t \le n$,*

$$a_t^{AS|j} \le a_t^{CS|j} \le a_t^{CAS|j} \le a_t^{AU|j}. \tag{1}$$

Consequently $m(AS,j) \le m(CS,j) \le m(CAS,j) \le m(AU,j)$, and the same holds for u, for every horizon law and every setting of the stage payoffs, the step sizes, the prize, $p_{\max}$ and the setback scope.

Proof. Write a design as a pair (f,c): a first action f and a flag c for whether it copies its opponent's previous action. Solving the two coupled recursions gives the focal path in closed form in each of three cases. If the opponent does not copy, it plays f_j in every round whatever the focal seat does, so a focal seat that does not copy plays f_i throughout, and one that copies plays f_i in round 1 and f_j afterwards. If both copy then $a_t = a_{t-2}$ for $t \ge 3$, so the focal seat plays f_i in odd rounds and f_j in even ones. In every case the path is either constant at f_i, or equal to f_i on a set of rounds that does not itself depend on f_i and to f_j on the complement.

AS plays Safe in every round and AU plays Unsafe in every round, against every opponent, so they are the pointwise smallest and largest paths in the design set and the outer two inequalities of (1) hold. CS and CAS carry the same flag $c = 1$ and differ only in f_i, which enters each closed form monotonically, so raising f_i from Safe to Unsafe raises the path in some rounds and lowers it in none, which gives the middle inequality.

At a fixed horizon, m counts the Unsafe entries of the path and u divides that count by n, so both are non-decreasing functions of the path and both inherit (1). Neither depends on any other parameter of the race, so taking the expectation over any law for T preserves the ordering. □

The same case analysis evaluates m exactly rather than only ordering it. With $\mu = \mathbb{E}[T]$ and $\pi = \Pr[T \text{ odd}]$, the rows of m in the erosion order are $(0,0,0,0)$ for AS, $\left(0, 0, \frac{\mu-\pi}{2}, \mu-1\right)$ for CS, $\left(1, \frac{\mu+\pi}{2}, \mu, \mu\right)$ for CAS and (μ,μ,μ,μ) for AU. At our horizon law $\mu = 9$ and $\pi = 5/9$, which reproduces Table 1 exactly. Every entry is bounded by μ, and that bound is the saturation Theorem 4 needs.

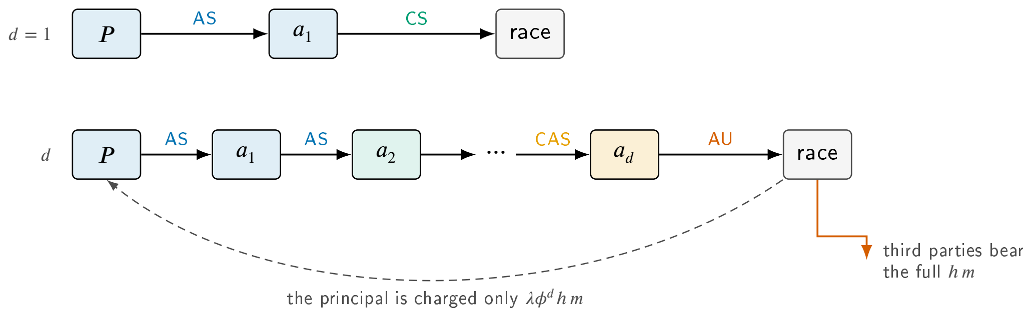

Figure 1: A delegation cascade. A principal P issues one specification and never acts. Each hand-off may drop a safety clause, so the design that reaches the race is a random function of the intent and of the depth alone; the deeper the chain, the further the executed design has slid down the erosion order AS → CS → CAS → AU. The harm the chain causes falls on third parties in full, but is charged back to the principal at the rate ϕ^d. Both effects are per-layer, so both are geometric in the depth, and they pull the principal in opposite directions: erosion makes depth expensive, attenuated attribution makes it cheap.

3.2. The delegation chain

A seat is operated not by a principal but by a chain

$$\text{principal} \longrightarrow \text{agent}_1 \longrightarrow \cdots \longrightarrow \text{agent}_d \longrightarrow \text{the race},$$

where $d \geq 0$ is the *delegation depth*. The principal issues one specification, its *intent* s, and each hand-off transmits it through the row-stochastic kernel

$$M = (1-\varepsilon)\, I + \varepsilon\, D, \tag{2}$$

with ε the per-hand-off erosion probability and D a drift kernel on the four designs. The identity term in (2) is the part of a specification that survives a hand-off untouched. Our baseline D is the *ladder*: an erosion event moves the specification one rung down the order AS, CS, CAS, AU, which is to say it drops one safety clause and leaves the description of the task intact. Two alternatives serve as controls. Under *collapse* an erosion event discards the specification and the sub-agent falls back on the locally competitive design AU. Under *uniform* it replaces the specification with a uniformly random one; that kernel has the same spectral gap [50] as the ladder but no direction, and it separates the effect of biased erosion from the effect of noise as such.

The executed design of a depth-d chain with intent s is therefore distributed as row s of M^d, written $p_{d,s}$. Figure 1 shows the construction and Figure 2 shows what it does to an AS instruction. We also allow one *check* in the chain: a check of strength α at layer $k \leq d$ restores the intent with probability α, after which the specification still has $d-k$ hand-offs to survive, so the executed design is distributed as $\alpha M^{d-k}(s,\cdot) + (1-\alpha) M^d(s,\cdot)$.

3.3. Private and social functionals

Two designs meeting in the race produce, in expectation over the executed designs and over the horizon,

$$a(d,s;d',s') = p_{d,s}^{\top} A\, p_{d',s'} + B(d), \qquad m(d,s;d',s') = p_{d,s}^{\top} M_h\, p_{d',s'}, \tag{3}$$

where A and M_h are the task-payoff and unsafe-action matrices of the interaction layer and $B(d) = g\,d - c\,d^2$ is the net organisational benefit of running a chain of depth d. Everything is bilinear because one specification is transmitted per race and then played, so the executed designs of the two seats are independent draws. The benefit B is real, in the sense that delegation genuinely produces more, and the term enters the social accounting as well as the private one. The baseline sets $g = 1.5$, about 2.5% of the payoff of a safe seat per hand-off, and $c = 0$, so that the only force limiting depth is the harm the principal internalises.

Let h be the external harm of one Unsafe action and λ the liability rate at depth zero. The *private* functional, which alone drives the dynamics, and the *social* functional, which alone scores them, are

$$\pi_P(d,s;d',s') = a(d,s;d',s') - \lambda\,\phi^d\, h\, m(d,s;d',s'), \tag{4}$$

$$\pi_S(d, s; d', s') = a(d, s; d', s') - h\, m(d, s; d', s'). \tag{5}$$

The two functionals differ only in the second term, and their difference is the *dilution wedge*

$$\Delta(d) \;=\; \pi_P - \pi_S \;=\; h\bigl(1 - \lambda\phi^d\bigr)\, m, \tag{6}$$

which is non-decreasing in d for every $\phi \leq 1$ and vanishes identically at $\phi = \lambda = 1$. Depth alone widens the gap between what a principal optimises and what society bears, without any change in what the principal intends. Only the product $L = \lambda h$ enters π_P, so the two-parameter family (λ, h) collapses to the single control L, the *effective liability*; h is a property of the world and stays fixed when the instrument moves, which is what keeps social payoffs comparable across liability regimes. Keeping the two functionals apart is not bookkeeping. Once the cost of an intervention is counted, driving a behavioural outcome to its maximum and maximising total population payoff are different objectives, and it is the second an intervention should be designed against [51]; we therefore report the long-run unsafe frequency and the social payoff separately throughout and never combine them.

3.4. Dynamics and the choice of parameters

Three timescales are separated, and the separation is an assumption rather than a derivation. A race runs to its horizon, $\mathbb{E}[T] = 9$ rounds, on the fastest; a chain is built once per race and transmits one specification, with no renegotiation while the race is under way; and designs are revised on the slowest, so that a principal reconsiders how deeply to delegate only between races and never inside one. The separation is what makes the executed designs of the two seats independent draws and the functionals bilinear. We take revision opportunities to arrive as a Poisson process, the standard assumption behind the pairwise-comparison dynamics used here. Kara et al. [52] show that when a strategy is decomposed into sub-tasks the inter-revision intervals become Erlang rather than exponential and the resulting population dynamics are not those of the memoryless case; a delegation chain is exactly such a decomposition, so the assumption is not innocuous and we return to it in Section 7.1.

Designs are copied in proportion to how well they pay. We report the stationary regime of the finite-population pairwise-comparison process [44] in the small-mutation limit [45–47] as the primary reading, with population size $Z = 100$ and selection intensity $\beta = 0.05$, and the basin-averaged attractor of the replicator equation as a secondary one.

The remaining choice is the liability level, and we fix it so that the delegation results cannot be an artefact of a race that was unsafe to begin with. Bisection on the depth-zero game puts the long-run unsafe frequency below 10^{-3} for every $L \geq 6.52$; we run at $L = 20$, that is 3.1 times the level at which the single-layer race is already safe. At that level the depth-zero ecosystem has long-run unsafe frequency 0.000 under the finite-population reading and 0.026 under the replicator reading. The baseline erosion probability is $\varepsilon = 0.20$ and the baseline attribution retention is $\phi = 0.75$; both are swept over their full ranges in Section 6.2 and Section 6.6.

4. Analytical results

4.1. The specification is forgotten geometrically

Under the ladder kernel the number of clauses lost over d hand-offs is binomial, so the executed design is the intent shifted down the erosion order by $\mathrm{Bin}(d, \varepsilon)$ rungs and truncated at AU. Two consequences are exact.

Proposition 2 (Erosion law). *For any intent other than the absorbing one, the specification survives d hand-offs intact with probability exactly* $(1 - \varepsilon)^d$*, and the influence of the intent on the executed design decays at the rate given by the second-largest eigenvalue modulus of* M*, which for the ladder and collapse kernels equals* $1 - \varepsilon$*. The* specification half-life $d_{1/2} = \ln 2 / \ln\bigl(1/(1-\varepsilon)\bigr)$ *therefore depends on nothing but the per-hand-off erosion probability.*

At the baseline $\varepsilon = 0.20$ the half-life is 3.11 hand-offs and only 26.2% of specifications reach a six-deep executor unaltered (Figure 2B,C). The claim is not that instructions are forgotten quickly; it is that they are forgotten at a rate that no amount of care at the top of the chain changes, because the decay is in the number of hand-offs and not in the effort spent writing the instruction.

Because the ladder kernel only moves mass down the erosion order, and because by Lemma 1 harm is monotone along that order, the harm a chain causes is monotone in its depth.

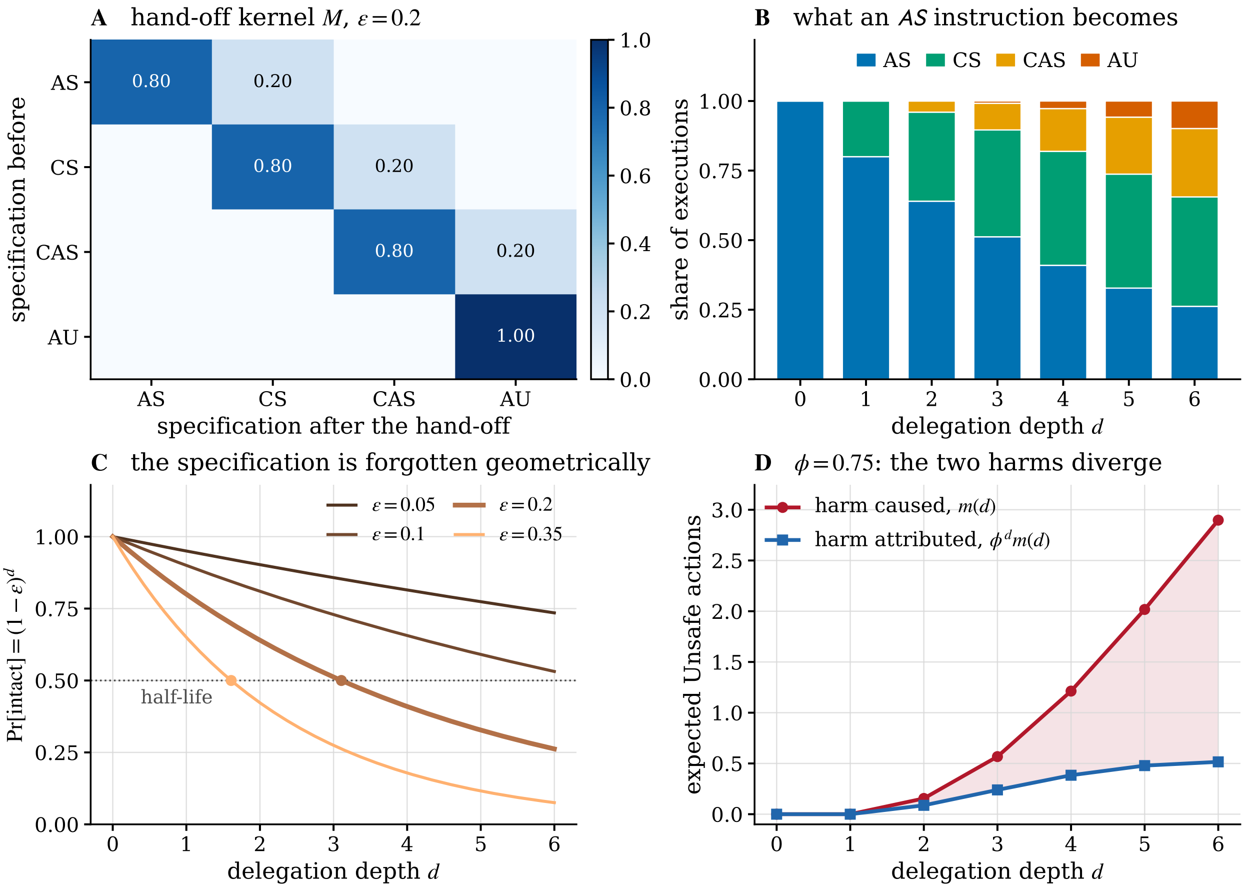

Figure 2: What a hand-off does to a specification. **(A)** The hand-off kernel at the baseline erosion probability: one clause is lost per erosion event, and the unconditionally unsafe design is absorbing. **(B)** The executed design of a chain that was told AS, by depth. At six hand-offs the instruction is executed as issued in 26% of cases and has slid to CAS or AU in 34%. **(C)** The probability that a specification arrives intact is exactly $(1-\varepsilon)^d$; dots mark the half-life, 3.11 hand-offs at $\varepsilon = 0.20$. **(D)** Harm caused and harm attributed for the same chain. The shaded wedge is the part of the damage that never reaches the principal's accounts.

Proposition 3 (Depth monotonicity). *Let M be a hand-off kernel that is stochastically monotone and satisfies $\delta_s \preceq M(s,\cdot)$ for every s in the erosion order. Then for every intent s and every opponent mixture, $d \mapsto m(d,s;\cdot)$ is non-decreasing. In particular the ladder and collapse kernels satisfy the hypothesis; the uniform kernel does not.*

Proof. $\delta_s \preceq M(s,\cdot)$ and stochastic monotonicity of M give $M^d(s,\cdot) \preceq M^{d+1}(s,\cdot)$ in the usual stochastic order by induction. Lemma 1 makes $m(\cdot,j)$ a non-decreasing function on the order for every j, and the expectation of a non-decreasing function respects the stochastic order. Bilinearity in (3) extends the statement from pure opponents to mixtures. □

4.2. Depth is a liability shelter

The harm a chain causes is bounded: once the specification has been forgotten, the executed design is the absorbing one whatever the intent, so $m(d)$ converges. The harm a chain is charged for carries the extra factor ϕ per layer and is bounded below by nothing. Comparing the two gives the central structural result of the model.

Theorem 4 (Liability shelter). *Adding a hand-off lowers the attributed harm exactly when*

$$\phi < \phi^*(d) = \frac{m(d)}{m(d+1)}. \tag{7}$$

Because m is non-decreasing and bounded, $\phi^*(d) \to 1$. Hence for every $\phi < 1$ there exists a depth $\bar{d}(\phi)$ such that every hand-off beyond $\bar{d}$ strictly reduces the harm attributed to the principal while strictly increasing the harm it causes, and

$$\lim_{d\to\infty} \phi^d m(d) = 0, \qquad \lim_{d\to\infty} m(d) = m_{\max}.$$

Probing (7) to depth 40 puts ϕ^* at 0.99993 and, at the baseline $\phi = 0.75$, the shelter opening at $\bar{d} = 6$ (Figure 3A,B). The theorem is the reason the model has no interior solution to appeal to: a principal that can add layers can drive its attributed harm to zero while driving its real harm to the maximum, and the only things that stop it are a rule limiting the number of layers or an attribution rule with $\phi = 1$ exactly.

The consequence for the liability instrument is sharp. In a resident population of depth-d principals with intent s, the advantage of a depth-$(d + 1)$ mutant is affine in the effective liability,

$$\pi_P(d{+}1, d) - \pi_P(d, d) \;=\; \underbrace{a(d{+}1, d) - a(d, d)}_{\text{benefit}} \;-\; L\,\underbrace{\left[\phi^{d+1} m(d{+}1, d) - \phi^d m(d, d)\right]}_{\text{attributed gain}}, \tag{8}$$

so the single ratio implied by (8) decides the outcome.

Proposition 5 (Invasion by depth). *The critical effective liability at which depth $d + 1$ loses its advantage over depth d is $L^*(d) =$ benefit/attributed gain. When the attributed gain is positive, $L > L^*(d)$ blocks the deeper design. When the attributed gain is negative, which by Theorem 4 happens for all d large enough whenever $\phi < 1$, the advantage is increasing in L and no level of liability blocks the deeper design.*

Figure 3C shows the attributed gain for each intent at the baseline. For the intent CAS it is negative at every depth, so for those principals liability is not a tax on delegation but a subsidy for it.

Corollary 6 (When a rule calibrated at depth zero stops binding). *A liability L that clears a threshold L_c for principals that act for themselves stops clearing it at depth $d^\dagger = \log(L_c/L)/\log\phi$, which is finite for every $\phi < 1$. At the baseline, $L_c = 6.52$, $L = 20$ and $\phi = 0.75$ give $d^\dagger = 3.9$: the rule survives three hand-offs and fails at the fourth.*

The last panel of Figure 3 records a fact that is easy to miss. The depth the population reaches is not the depth that is best for the principals themselves. A monomorphic population of one-hand-off principals earns more than a monomorphic population of six-hand-off principals, and yet the deeper design invades at every step, because an invader is judged against the resident environment rather than against its own. Delegation depth is a social dilemma at the private level before it is one at the social level.

4.3. Where a check belongs, and what closes the shelter

Two further statements are exact and both bear directly on the instruments compared in Section 6.4. The first orders the placements of a single check along the chain.

Proposition 7 (Audit placement). *Under a kernel satisfying the hypothesis of Proposition 3, the realised harm of a depth-d chain carrying one check of strength α at layer k is non-increasing in k.*

Proof. The executed law is $\alpha M^{d-k}(s,\cdot) + (1-\alpha)M^d(s,\cdot)$, and by Proposition 3 the expected harm under M^j is non-decreasing in j. Increasing k decreases $d - k$, hence decreases the first term and leaves the second unchanged. $\square$

The second identifies what an attribution rule has to violate for Theorem 4 to lose its force. The theorem needs only $a(d)m(d) \to 0$, where $a(d)$ is the share of the realised harm charged to the principal; the geometric rule $a(d) = \phi^d$ is one way of arranging that, and so is any rule that keeps falling.

Proposition 8 (A floor closes the shelter). *Let the attribution rule satisfy $a(d) = a_{\min} > 0$ for every $d \ge d_0$, and let the hand-off kernel satisfy the hypothesis of Proposition 3. Then $d \mapsto a(d)\,m(d)$ is non-decreasing on $d \ge d_0$, so the private gain from one further hand-off there is at most $B(d+1) - B(d)$ and no depth beyond d_0 is a shelter in the sense of Theorem 4.*

Proof. For $d \ge d_0$, $a(d+1)m(d+1) - a(d)m(d) = a_{\min}\left[m(d+1) - m(d)\right] \ge 0$ by Proposition 3. The attributed-harm term of π_P is therefore non-increasing in depth beyond d_0, leaving the organisational benefit as the only remaining gain. $\square$

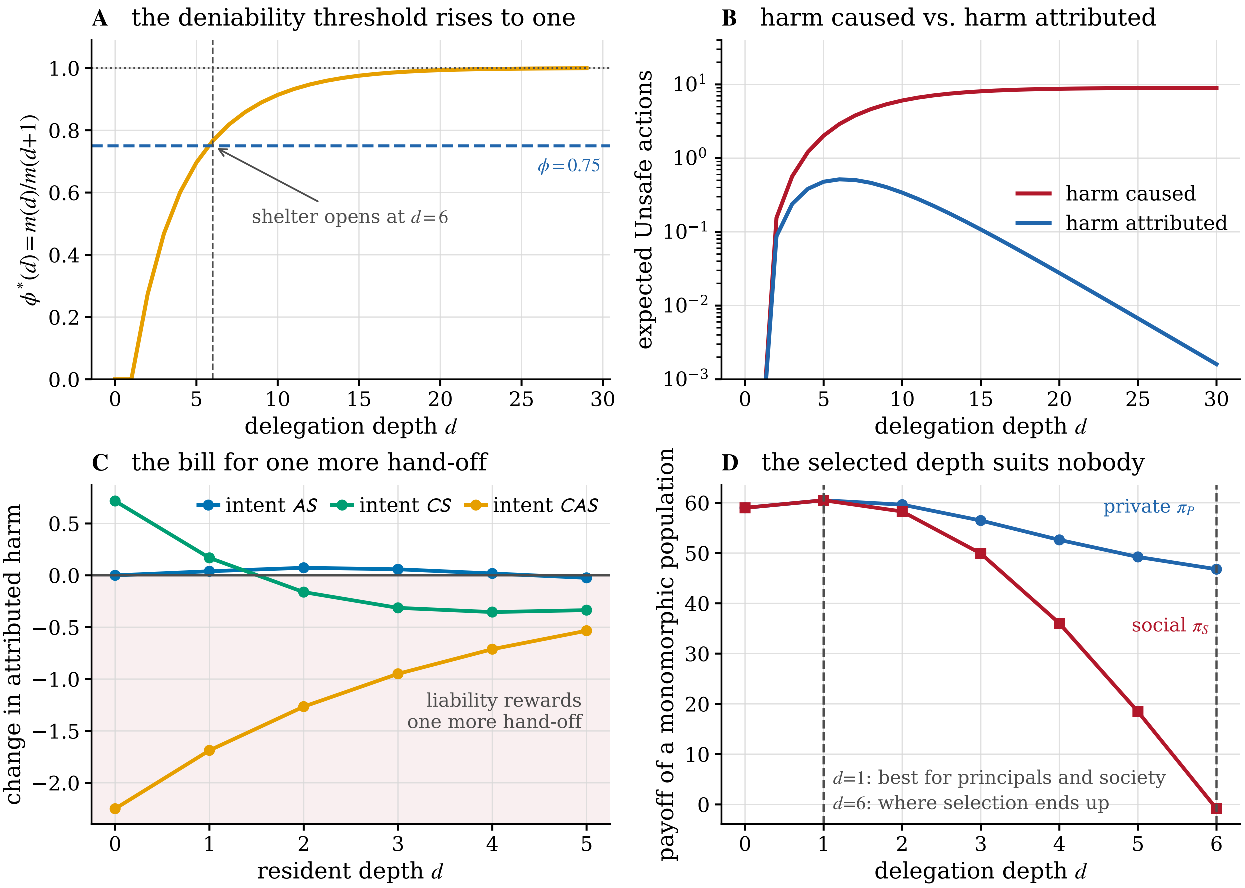

Figure 3: Depth as a liability shelter. **(A)** The deniability threshold $\phi^*(d) = m(d)/m(d+1)$ rises towards one because realised harm saturates, so any attribution retention below one is eventually below the threshold; at $\phi = 0.75$ the shelter opens at six hand-offs. **(B)** Beyond that depth, realised harm still climbs towards its ceiling while attributed harm falls geometrically. **(C)** What one more hand-off adds to the principal's bill. Where the curve is negative, liability rewards delegation instead of taxing it. **(D)** Private and social payoff of a monomorphic population by depth. Selection ends at the ceiling, which is worse for the principals than one hand-off and much worse for everybody else.

5. Numerical method

Everything above is exact given the two matrices A and M_h and the transmission laws $p_{d,s}$. What is not immediate is the stationary regime of the evolutionary process on the joint design space, which is the object every number in Section 6 is a functional of. This section sets out how it is computed, at what cost, and against what it is checked.

5.1. Exact evaluation of the interaction tensor

The four reduced designs are deterministic, so for a fixed ordered pair (i, j) the two action paths are fixed and the only randomness in a race is its horizon T. Truncating the horizon law at $T_{\max} = 400$, with the residual mass $(1 - 0.2)^{T_{\max}-5}$ assigned to $T_{\max}$ so that the probabilities sum to one exactly, the expected task payoff and expected unsafe count of i against j are finite sums

$$a(i,j) = \sum_t \Pr[T = t]\, a_t(i,j), \quad m(i,j) = \sum_t \Pr[T = t]\, m_t(i,j), \quad u(i,j) = \sum_t \Pr[T = t]\, u_t(i,j), \tag{9}$$

in which a_t, m_t and u_t are read off prefix sums of the two action paths, u_t being the Unsafe count of the path divided by its length. We write A, M_h and U_h for the three matrices they define. All three are therefore obtained in $O(|S|^2 T_{\max})$ operations at machine precision, with $|S| = 4$, and carry no sampling error at all. The alternative, simulating races and

Table 2
Cost of the two routes to the stationary regime. The full mutation–selection process is a Markov chain on population states, of which there are $\binom{Z+n-1}{n-1}$; the reduction of Section 5.2 needs $n(n-1)$ fixation probabilities, each evaluated in $O(Z)$ arithmetic operations. The depth ceiling $\bar{D}$ fixes the number of designs at $n = 4(\bar{D}+1)$; the manuscript runs at $\bar{D} = 6$, $Z = 100$.

$\bar{D}$	n	Z	states of the full chain	operations of the reduction
4	20	50	4.6×10^{16}	19 000
4	20	100	4.9×10^{21}	38 000
4	20	200	1.1×10^{27}	76 000
6	28	50	4.4×10^{20}	37 800
6	28	100	3.0×10^{27}	75 600
6	28	200	7.5×10^{34}	151 200
8	36	50	9.0×10^{23}	63 000
8	36	100	2.8×10^{32}	126 000
8	36	200	6.5×10^{41}	252 000

averaging, is what the size of the sweeps below makes unattractive rather than merely inelegant: the root-mean-square error of the payoff matrix is still 0.034 after 10^6 episodes per matchup (Table 3), and every one of the 1764 stationary regimes behind the parameter plane of Section 6.2 would inherit it.

5.2. Why the full mutation–selection chain is out of reach

Under the pairwise-comparison rule with mutation rate μ, the population of Z principals is a Markov chain whose state is the vector of counts of each of the $n = 4(\bar{D}+1)$ designs. The number of such states is $\binom{Z+n-1}{n-1}$, and Table 2 evaluates it. At the baseline $\bar{D} = 6$ and $Z = 100$ it is 3.0×10^{27}, so neither the transition matrix nor its stationary vector can be formed, and raising the ceiling to $\bar{D} = 8$ or the population to $Z = 200$ makes it worse by five and seven orders of magnitude respectively. Any study that puts a structural variable such as depth under selection meets this wall immediately, because the design space is a product of the strategy space with the range of the structural variable.

We therefore work in the small-mutation limit [45, 47], in which a mutation either fixes or is lost before the next arrives, the population is monomorphic almost always, and the process reduces to an embedded chain on the n pure designs with

$$P_{ji} = \frac{\rho_{i\to j}}{n-1} \quad (j \neq i), \qquad P_{ii} = 1 - \sum_{j\neq i} P_{ji}, \tag{10}$$

where $\rho_{i\to j}$ is the probability that a single j mutant takes over a resident population of i. The state space has fallen from $\binom{Z+n-1}{n-1}$ to n. What the reduction costs is the mutation rate: it is exact only in the limit $\mu \to 0$, and Section 5.6 measures the error at finite μ against the full chain on a design space small enough to carry one. How small a rate the embedded chain needs has been characterised analytically for imitation processes at arbitrary selection intensity [48], which is the class our update rule belongs to.

5.3. Fixation probabilities in closed form, evaluated stably

Let A denote the private payoff matrix π_P over designs, and consider a population holding k copies of an invading design I and $Z-k$ of a resident R. Sampling an opponent without replacement, the expected payoffs are

$$\pi_I(k) = \frac{k-1}{Z-1}A_{II} + \frac{Z-k}{Z-1}A_{IR}, \qquad \pi_R(k) = \frac{k}{Z-1}A_{RI} + \frac{Z-k-1}{Z-1}A_{RR}. \tag{11}$$

Under the pairwise-comparison rule with Fermi function and selection intensity β, an individual switches from R to I with probability $[1+e^{-\beta(\pi_I-\pi_R)}]^{-1}$, so the birth-death chain on k has $T^-(k)/T^+(k) = e^{-\beta(\pi_I(k)-\pi_R(k))}$ and the standard formula for a one-dimensional birth-death chain with absorbing ends gives, as Appendix A derives,

$$\rho_{R\to I} = \left[1 + \sum_{m=1}^{Z-1} \exp\!\Big(-\beta \sum_{k=1}^{m} \big(\pi_I(k) - \pi_R(k)\big)\Big)\right]^{-1}. \tag{12}$$

Algorithm 1 Stationary regime of the delegation game

Require: race parameters, ceiling $\bar{D}$, erosion ε, kernel D, attribution rule $a(\cdot)$, liability L, harm h, Z, β, and optionally one check (α, k)

Ensure: x, and the observables U, $\bar{d}$, π_S, declaration gap

1: $A, M_h, U_h \leftarrow$ exact horizon sums (9) $\triangleright\ O(|S|^2 T_{\max})$

2: $M \leftarrow (1-\varepsilon)I + \varepsilon D$; $\quad p_{d,\cdot} \leftarrow M^d$, or $\alpha M^{d-k} + (1-\alpha)M^d$ with a check, for $d = 0, \dots, \bar{D}$ $\triangleright\ O(\bar{D}|S|^3)$

3: assemble π_P, π_S, u on the $n = 4(\bar{D}+1)$ designs by (3)–(5) $\triangleright\ O(n^2|S|^2)$

4: **for** each ordered pair (i, j), $i \neq j$ **do**

5: $\quad\theta \leftarrow$ cumulative sum of $-\beta(\pi_j(k) - \pi_i(k))$, $k = 1, \dots, Z-1$ $\triangleright\ O(Z)$

6: $\quad\rho_{i\to j} \leftarrow$ shifted sum (13)

7: **end for**

8: build P by (10); $\quad x \leftarrow$ left Perron vector of P $\triangleright\ O(n^3)$

9: **return** x and its functionals

Because (11) is affine in k, the inner sum is a quadratic polynomial in m, the whole vector of $Z-1$ exponents is one cumulative sum, and (12) is evaluated exactly in $O(Z)$ operations with no simulation. Evaluating it *as written* is another matter. The exponents grow like $\beta Z \max_k |\pi_I(k) - \pi_R(k)|$, and the payoff spread of this game is wide because a design that meets an eroded opponent can lose the whole prize. Over the 756 ordered pairs of the baseline design space the largest exponent reaches 860.8 at $\beta = 0.05$ and 3443.2 at $\beta = 0.2$, against an overflow threshold of $\ln(\text{DBL\_MAX}) = 709.8$: the naive sum overflows on 18 of the 756 pairs at the baseline selection intensity and on 167 of them at the strongest intensity in our sweeps, returning $\rho = 1/(1+\infty) = 0$ where the true value is positive. We therefore factor the largest exponent out before summing,

$$\rho_{R\to I} = \frac{e^{-\sigma}}{e^{-\sigma} + \sum_m e^{\theta_m - \sigma}}, \qquad \sigma = \max\Big(0,\ \max_m \theta_m\Big), \tag{13}$$

with θ_m the exponent of the m-th term of (12). The shift is exact in real arithmetic and keeps every exponentiated quantity in $[0, 1]$. It recovers a strictly positive fixation probability on six of the pairs the naive sum loses at each selection intensity, the smallest of them 2×10^{-323}, and Section 5.6 shows that it costs no accuracy where the naive sum survives.

5.4. The embedded chain and the reported observables

The stationary distribution x of (10) is its left Perron vector, and it is unique: the embedded chain is irreducible whenever every $\rho_{i\to j}$ is strictly positive, which Appendix A shows is exactly what the shifted evaluation (13) preserves and the naive sum destroys. We obtain it by replacing the last equation of the singular system $(P^\top - I)x = 0$ with the normalisation $\mathbf{1}^\top x = 1$ and solving the resulting nonsingular system directly, falling back on power iteration when the direct solve is ill conditioned. Every population quantity reported below is a functional of x: the long-run unsafe frequency is $U = \sum_{ij} x_i x_j u(i, j)$, the mean delegation depth is $\bar{d} = \sum_i x_i d_i$, the social payoff is $\sum_{ij} x_i x_j \pi_S(i, j)$, and the *declaration gap* is the total variation distance between the intent marginal of x and the executed-design marginal $x^\top P_{\text{tr}}$, where P_{tr} stacks the transmission laws $p_{d,s}$. Nothing here is estimated. Algorithm 1 collects the whole pipeline.

5.5. Cost

Steps 4 to 7 of Algorithm 1 dominate, at $n(n-1)$ fixation probabilities of $O(Z)$ each, so one stationary regime costs $O\big(n^2 Z + n^3\big)$ arithmetic operations against the $\binom{Z+n-1}{n-1}$ states of the route it replaces. At the baseline that is 75 600 operations for the fixation stage and 28 milliseconds end to end in a NumPy implementation, so the 21×21 grid of Section 6.2, which needs four stationary regimes per grid point, completes in under a minute. The stationary solve is well behaved: the rows of P sum to one to 2.2×10^{-16} and the residual $\|x^\top P - x^\top\|_\infty$ of the returned vector is 4.2×10^{-17}.

5.6. Verification

Table 3 reports four independent checks.

Against sampling. A Monte-Carlo estimate of the same payoff matrix converges at the expected $N^{-1/2}$ and is still two decimal places short of the exact value at 10^6 episodes per matchup. The exact route removes that error and the

Table 3
Verification of the scheme, in four independent checks. Block 1 compares the exact horizon expectation of Section 3.1 with a Monte-Carlo estimate of the same payoff matrix: root-mean-square error over the sixteen ordered pairs and eight independent replicates, which decays at the Monte-Carlo rate $N^{-1/2}$ and which the exact route removes. Block 2 compares the stabilised evaluation of (12) in double precision with the same sum evaluated in 60-digit arithmetic, worst relative error over the $n(n-1) = 756$ ordered pairs of the full design space whose fixation probability is representable as a normal double. Block 3 compares the closed form with the general-purpose routine of EGTtools on the depth-zero design space, worst absolute difference over all ordered pairs. Block 4 compares the small-mutation limit with the full mutation–selection chain on a design space small enough for it ($n = 3$, $Z = 40$, 861 population states).

check	setting	discrepancy
Monte Carlo vs. exact	$N = 1\,000$ episodes	1.080
Monte Carlo vs. exact	$N = 10\,000$ episodes	0.269
Monte Carlo vs. exact	$N = 100\,000$ episodes	0.117
Monte Carlo vs. exact	$N = 10^6$ episodes	0.034
stabilised vs. 60-digit sum	$n = 28,\ Z = 100,\ \beta = 0.05$	4.4×10^{-16}
stabilised vs. 60-digit sum	$n = 28,\ Z = 100,\ \beta = 0.2$	2.8×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 50,\ \beta = 0.05$	1.1×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 50,\ \beta = 0.2$	1.1×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 100,\ \beta = 0.05$	2.8×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 100,\ \beta = 0.2$	5.6×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 200,\ \beta = 0.05$	1.1×10^{-16}
closed form vs. EGTtools	$n = 4,\ Z = 200,\ \beta = 0.2$	1.1×10^{-16}
small mutation vs. full chain	$\mu = 5 \times 10^{-2}$	0.0910
small mutation vs. full chain	$\mu = 2 \times 10^{-2}$	0.0426
small mutation vs. full chain	$\mu = 10^{-2}$	0.0225
small mutation vs. full chain	$\mu = 5 \times 10^{-3}$	0.0116

sweeps below are correspondingly free of it, which matters because the interaction index of Section 6.2 is a difference of four stationary regimes and would otherwise be a difference of four noisy ones.

Against a high-precision reference. Evaluating the sum in (12) in 60-digit arithmetic and comparing with the shifted double-precision evaluation (13) over all 756 ordered pairs gives a worst relative error of 4.4×10^{-16} at $\beta = 0.05$ and 2.8×10^{-16} at $\beta = 0.2$, taken over the pairs whose fixation probability is representable as a normal double. The shift therefore costs nothing where the naive sum survives and recovers the value where it does not.

Against an independent implementation. On the depth-zero design space, which is small enough for a general-purpose routine organised around the full state space, the closed form agrees with EGTtools [49] to at most 5.6×10^{-16} at every population size and selection intensity tried. The one systematic difference is in the other direction: the general-purpose routine returns exactly zero for strongly disadvantaged mutants that (13) still resolves.

Against the full chain. On a design space of three designs, where the full mutation–selection chain has 861 population states and can be solved directly, the small-mutation limit is approached monotonically as μ falls, with an absolute difference in long-run unsafe frequency of 0.0116 at $\mu = 0.005$. That residual is the price of the reduction, and it is small against the effects reported below, which are of order 0.3.

Against the alternative dynamics. Every headline number is additionally recomputed under the basin-averaged attractor of the replicator equation, an entirely different limit of the same game, and reported alongside the finite-population value in Section 6.1.

6. Numerical results

6.1. Neither mechanism is dangerous alone

The two mechanisms can be switched off independently: $\varepsilon = 0$ removes erosion and $\phi = 1$ removes attenuated attribution. Crossing them gives a 2×2 design whose four cells are reported in Table 4 and Figure 4.

Starting from a race that is already safe, erosion alone raises the long-run unsafe frequency to 0.018 and attenuated attribution alone to 0.012. An additive reading predicts 0.030 when both are present. The measured value is 0.322. The

Table 4
The two mechanisms, separately and together. Long-run Unsafe frequency U, mean delegation depth $\bar{d}$, declaration gap and social payoff in the stationary regime of the finite-population process. Neither mechanism does much alone; the interaction is 0.293, or 91% of the joint effect.

erosion ε	attribution ϕ	U	$\bar{d}$	gap	π_S
0	1	0.0000	6.00	0.000	68.0
0.2	1	0.0176	1.99	0.358	58.3
0	0.75	0.0115	5.95	0.000	65.5
0.2	0.75	0.3225	6.00	0.738	-0.8
interaction		+0.2934			

interaction, +0.293, is 91% of the joint effect, and the same picture appears under the replicator reading, where the interaction is +0.331.

The mechanism behind the interaction is visible in the second panel of Figure 4, and it is not what one would guess. Under full pass-through, erosion is a private cost, and the population responds to it by *shortening* its chains: mean depth falls from 6.00 to 1.99. Erosion is self-limiting when the principal is charged for it. Attenuated attribution does not make behaviour worse at a given depth; it removes the response. With $\phi = 0.75$ the chains stay at the ceiling and the same erosion now acts over six hand-offs instead of two.

A counterfactual makes the decomposition exact. Scoring the pass-through population with the per-layer harm matrix, the same designs charged differently, leaves the unsafe frequency at 0.018, unchanged to nine decimal places, because the behaviour of a given design does not depend on who pays for it. The whole of the remaining +0.305 is composition, that is, which designs the population comes to contain. The result is a caution about a natural inference. Observing that deeper chains behave worse does not license the conclusion that the harm comes from erosion, because in this model almost all of it comes from the depth the market selects once erosion is no longer its problem.

6.2. The attribution–erosion plane

Neither baseline value is doing the work. Figure 5 sweeps the plane (ε, ϕ) on a 21×21 grid. Writing $U(\varepsilon, \phi)$ for the long-run unsafe frequency of the stationary regime at that pair, the *interaction index* is the difference of differences of the 2×2 whose corners are the pair itself and the two switched-off mechanisms,

$$I(\varepsilon, \phi) \;=\; U(\varepsilon, \phi) - U(\varepsilon, 1) - U(0, \phi) + U(0, 1), \tag{14}$$

and the interaction share quoted alongside it is $I/\big[U(\varepsilon, \phi) - U(0, 1)\big]$, the fraction of the total rise that neither mechanism produces alone. Every point of the grid therefore costs four stationary-regime computations, one per corner, each exact given the payoff matrices rather than sampled; the whole figure is 1764 of them. The index is positive over 89% of the interior and reaches 0.81; the long-run unsafe frequency reaches 0.995 and the declaration gap 0.95.

The second panel is the one to read for policy. Mean delegation depth splits the plane along a frontier: below it the ceiling binds and chains are as long as the technology allows, above it chains are short. The frontier runs diagonally, so erosion and attribution are substitutes in producing long chains. A regime with faithful hand-offs tolerates much weaker attribution before the ceiling binds, and a regime with strong attribution tolerates much noisier hand-offs.

6.3. Declared safety and executed safety

At the baseline the population is monomorphic on the design AS@6: every principal issues the safest available instruction and every principal runs a six-hand-off chain. What is executed is another matter. The realised mixture is 26% AS, 39% CS, 25% CAS and 10% AU, the total variation distance between what is declared and what is executed is 0.74, and social payoff falls from 68.0 in the baseline cell to −0.8.

The equilibrium is easy to misread as bad faith, and it is not. Every principal is issuing the strongest safety instruction available to it, and doing so is a best response: over-specification is how a principal compensates for a chain that will erode whatever it is told. The classical route to an uninformative message runs through the sender, since in the cheap-talk benchmark a message conveys less the further the sender's preferences diverge from the receiver's and a wholly uninformative equilibrium always exists [53]. Nothing of that kind is at work here. The declaration is sincere and as strong as the language permits, and the information in it is destroyed downstream of the party that issued

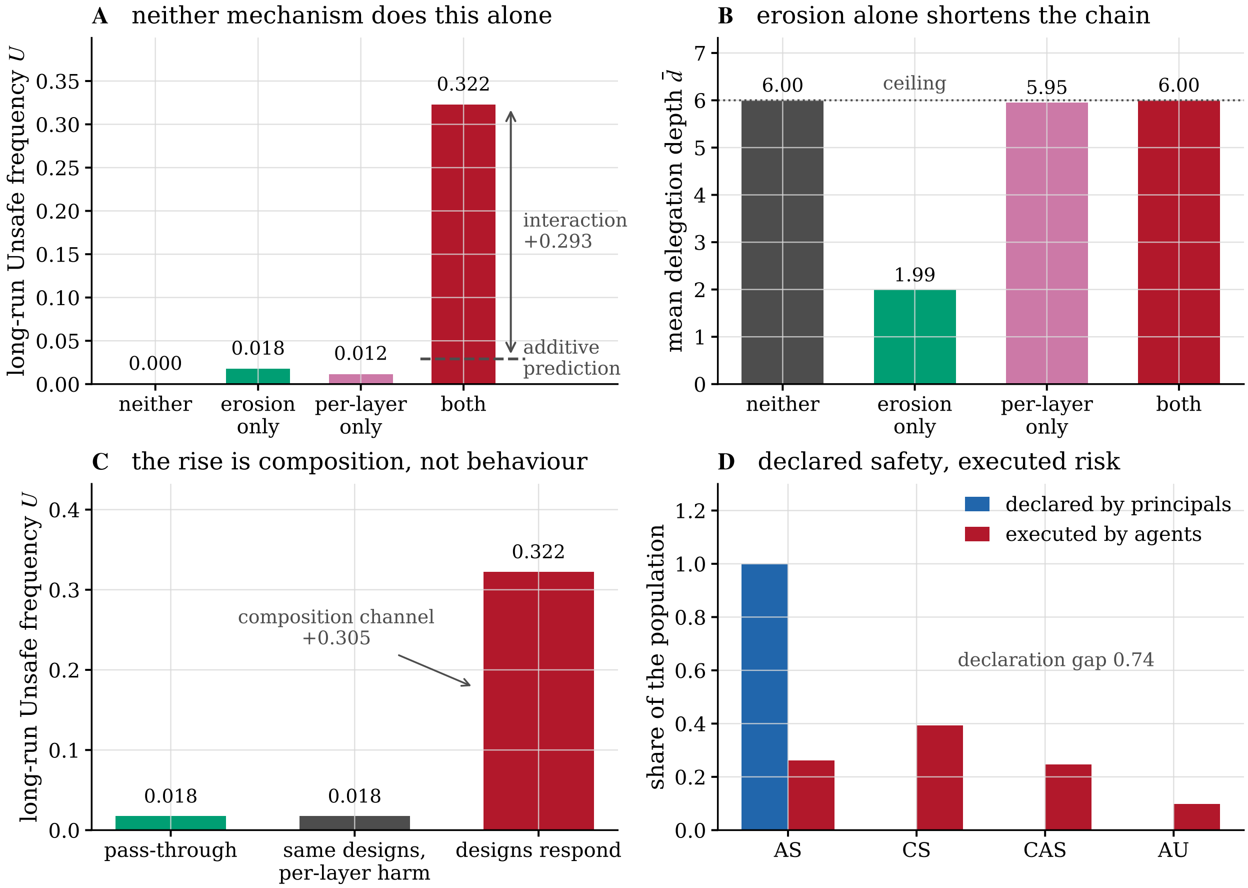

Figure 4: The two mechanisms, separately and together. **(A)** Long-run unsafe frequency in the four cells of the 2×2. The additive prediction from the two single-mechanism cells is 0.030; the joint outcome is 0.322. **(B)** Mean delegation depth in the same four cells. Under pass-through the population answers erosion by shortening its chains; per-layer attribution removes that answer and the ceiling binds. **(C)** Holding the pass-through population fixed and charging it the per-layer bill changes nothing, so the entire rise is composition rather than behaviour. **(D)** In the joint cell every principal declares the safest instruction and the population executes something else.

it. What the equilibrium shows is that a declaration made at the top of a long chain carries almost no information about the behaviour at the bottom of it.

6.4. What the instruments do

Four instruments act at different points of the chain: raise the attribution retention ϕ towards pass-through; cap the permitted chain length at $\bar{D}$; make every hand-off more faithful, multiplying ε by a factor κ; or insert one check of strength α at a chosen layer. Figure 6 applies each to the same baseline.

Attribution is a threshold, not a dial. Raising ϕ from 0.75 does nothing at all until $\phi = 0.86$, where mean depth falls from 5.92 to 3.45 and the unsafe frequency from 0.314 to 0.103; by $\phi = 0.99$ the outcome is the pass-through outcome. The location of the jump is close to the closed form implied by Corollary 6, $\phi_c = (L_c/L)^{1/\bar{D}} = 0.830$, the residual gap being the extra harm that erosion contributes at depth. Partial pass-through buys very little: at $\phi = 0.85$, which returns 85% of responsibility at each of six layers and therefore 38% overall, the ecosystem is still at 0.314.

A ceiling buys safety and welfare at once. The depth ceiling is the only instrument here that is monotone in its own strength and improves both objectives together. A ceiling of one hand-off removes the unsafe behaviour entirely

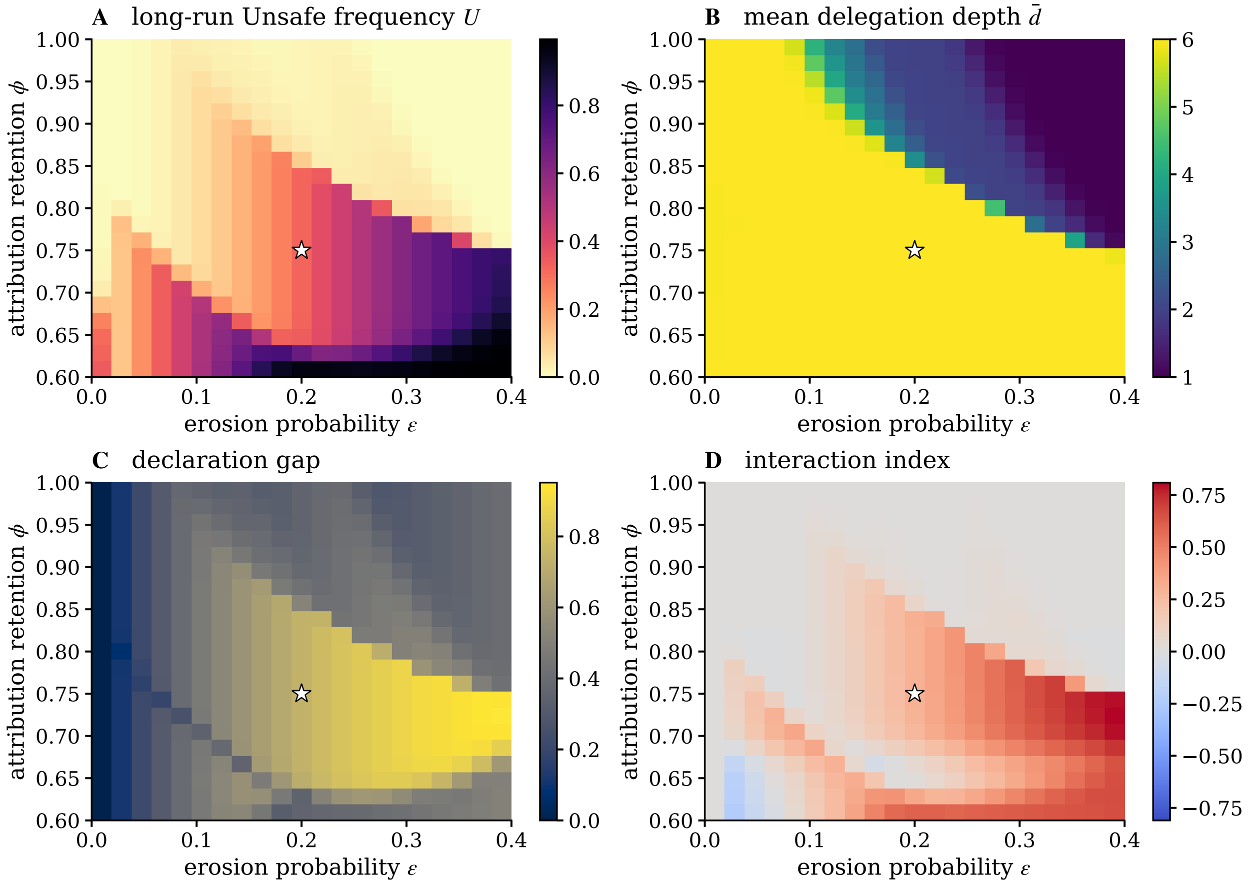

Figure 5: The attribution–erosion plane. Long-run unsafe frequency **(A)**, mean delegation depth **(B)**, declaration gap **(C)** and interaction index **(D)** over a 21×21 grid; the star marks the baseline. Chains are at the ceiling throughout the yellow region of **(B)**, and the frontier that bounds it is diagonal: attribution and transmission fidelity substitute for each other in keeping chains short.

and lifts social payoff from −0.8 to 60.5; a ceiling of two leaves 0.018 and 58.3. It works because the ceiling is the constraint the shelter theorem leaves standing when attribution cannot be made exact.

Fidelity is not monotone. Improving transmission helps over most of its range, from 0.322 at $\kappa = 1$ down to 0.032 at $\kappa = 0.35$, and then reverses: at $\kappa = 0.20$ the unsafe frequency is back up to 0.239. The cause is the intent margin. Over-specification was an equilibrium response to erosion; once hand-offs are faithful enough, issuing CS rather than AS becomes the better reply, and CS eroding into CAS is far more damaging than AS eroding into CS. A partial fidelity improvement can therefore leave the ecosystem worse off than no improvement at all. The structure is that of offsetting behaviour, in which a technological safety improvement displaces the private precaution that had been standing in for it [54, 55], and over-specification is the precaution our hand-off technology displaces.

The flip can be located exactly. Write the margin as $\pi_P(\mathsf{CS}@\bar{D}) - \pi_P(\mathsf{AS}@\bar{D})$ against a resident $\mathsf{AS}@\bar{D}$ population. It is zero at $\varepsilon = 0$, where the two intents are executed as issued and neither meets an unsafe opponent; it rises to a maximum of 4.00 at $\varepsilon = 0.04$; and it falls back through zero at $\varepsilon = 0.0975$, just under half the baseline erosion. Improving fidelity therefore does not move the ecosystem along a monotone curve but along a hump. The three consequences are separated in ε: the margin changes sign at 0.098, the unsafe frequency reaches its minimum of 0.031 at 0.065, and the population actually switches its declared intent from AS to CS at 0.050, after which the unsafe frequency rises to 0.239. The lag between the sign change and the switch is the fixation barrier of the finite population: a positive margin is necessary for the invasion, not sufficient.

Table 5
Attribution regimes. Long-run Unsafe frequency, mean delegation depth and social payoff when the law mapping depth to the principal's share of the harm is changed, with everything else at the baseline. The shelter survives every rule whose attribution keeps falling in the depth, including the equal split, and closes under every rule that is bounded below.

rule	setting	$a(D)$	U	$\bar{d}$	π_S
geometric, ϕ^d	$\phi = 0.75$	0.178	0.3225	6.00	-0.8
strict, 1	–	1.000	0.0176	1.99	58.3
equal split, $1/(1+d)$	–	0.143	0.3224	6.00	-0.8
floored, $\max(\phi^d, a_{\min})$	$a_{\min} = 0.25$	0.250	0.2317	5.07	17.1
floored, $\max(\phi^d, a_{\min})$	$a_{\min} = 0.5$	0.500	0.0633	2.99	50.0
super-attribution, ϕ^d	$\phi = 1.1$	1.772	0.0159	1.92	58.5
super-attribution, ϕ^d	$\phi = 1.25$	3.815	0.0030	1.21	60.2

The mechanism generalises beyond this particular race. It needs two things: an intent that is safe only conditionally and that outperforms the unconditionally safe intent against an unsafe opponent, which here is CS earning 26.6 against CAS where AS earns 8.6; and a resident population whose executed mixture contains unsafe designs for the conditional intent to exploit, which erosion itself supplies.

Check late, not early. Proposition 7 orders the placements for a fixed design, and in the equilibrium the effect is large: the same check is worth a factor of 18 more at the agent that acts than at the principal, 0.018 against 0.322. Checking the specification a party *received* tells one little about what will be done with it several hand-offs later. The sweep is non-monotone at full check strength for the same reason the fidelity sweep is, and monotone at half strength, where the intent margin does not flip.

Which instrument is cheapest. Asking each instrument for the same target, a long-run unsafe frequency of at most 0.05, and scoring it by the social payoff it delivers there (Figure 7B): improving fidelity reaches the target at 58.7, the depth ceiling at 58.3, a punitive liability at 56.2 and pass-through at 53.5. Comparing instruments by the welfare cost of holding a population outcome at a fixed level is the exercise Duong and Han [56] formalise for institutional incentives in a finite stochastic population, where which of two instruments is cheaper turns out to depend on the parameters rather than on the instrument. The ranking here is close enough that we would not press it, but the qualitative message is not: raising the liability level is the least efficient of the four, because it acts on a margin, namely how much harm costs, that depth has already discounted.

6.5. A floor under attribution

Geometric attenuation is a modelling choice, so the shelter could be an artefact of it and the instruments could be artefacts of the artefact. Neither is the case, and separating the two questions, how fast attribution falls and whether it falls at all, identifies an instrument the earlier comparison missed. Table 5 replaces ϕ^d by four alternative rules, changing nothing else.

Two of them reproduce the baseline almost exactly. Splitting the blame equally between the principal and its d agents, $a(d) = 1/(1+d)$, is a much slower decay than ϕ^d and it is the rule an intuitive notion of shared responsibility suggests; it gives an unsafe frequency of 0.3224 against the baseline 0.3225, the same mean depth of 6.00 and the same declaration gap. Top-level strict liability, $a(d) \equiv 1$, is the same object as pass-through and reproduces the $\phi = 1$ cell of Section 6.1. Super-attribution, $\phi > 1$, charges a principal more the deeper it delegates and closes the shelter more than completely, but it buys little over strict liability: 0.016 at $\phi = 1.1$ against 0.018 at $\phi = 1$.

What separates the rules is not the level of attribution but whether it keeps falling, which is exactly the content of Proposition 8. The equal split ends lower than the baseline at the ceiling ($1/7 = 0.143$ against $\phi^6 = 0.178$) and behaves identically, while a rule that stops falling behaves entirely differently even when it is lower in between. The mechanism is visible in the marginal attributed harm, the quantity a principal weighs when it considers one more layer. Under the baseline it rises and then decays, 0.087, 0.152, 0.144, 0.095, 0.037, so the sixth hand-off is nearly free. Under a floor at $a_{\min} = 0.5$ it climbs throughout, 0.087, 0.196, 0.323, 0.402, 0.440, so every layer costs more than the one before it.

The practical consequence is the reason to report this at all. A floor is not a weak version of pass-through, it is a substitute for a depth cap. Table 6 matches each ceiling against the floor delivering the same social payoff and finds it

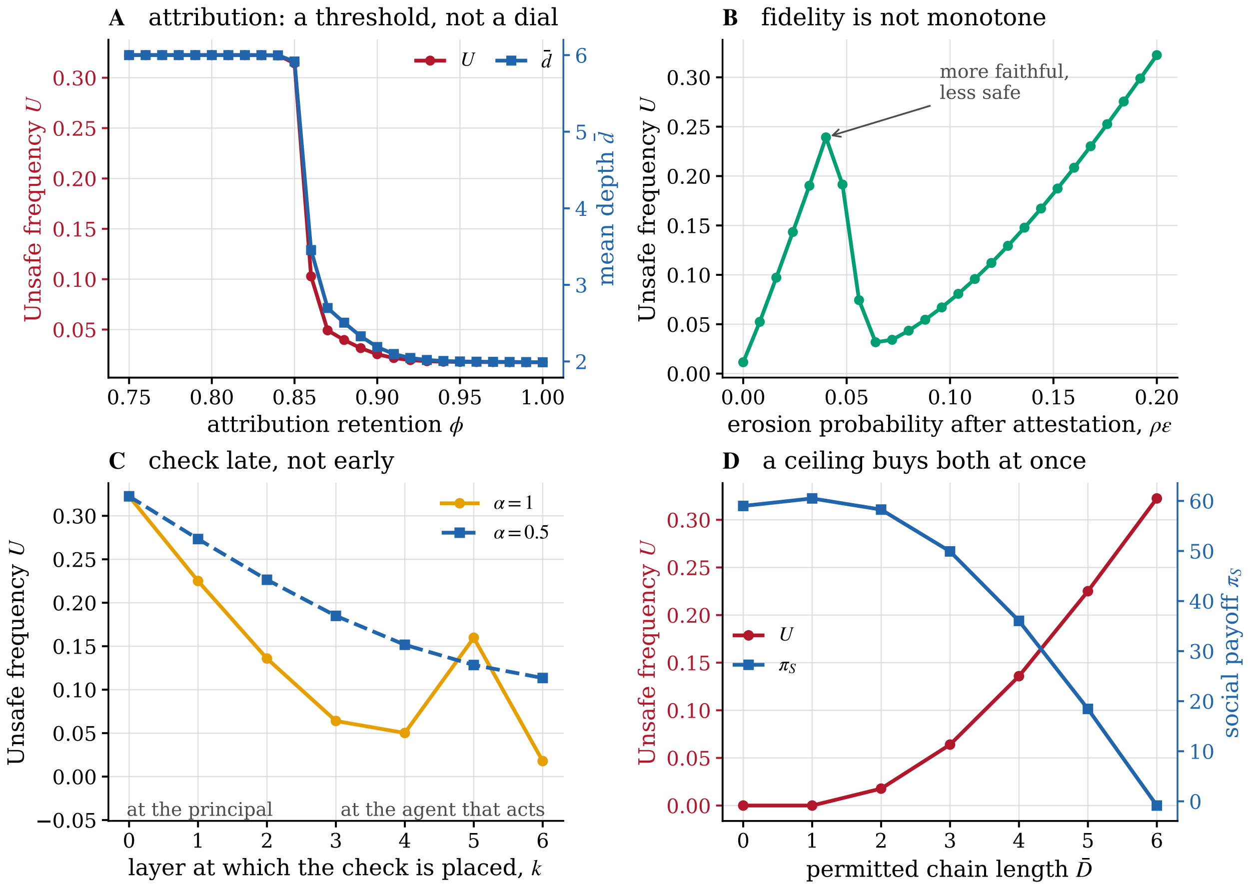

Figure 6: Four instruments. **(A)** Restoring attribution has a threshold at $\phi = 0.86$ and does almost nothing below it. **(B)** Making every hand-off more faithful is not monotone: the marked point is safer than settings with less erosion. **(C)** One check, moved down the chain. Late placement dominates; the non-monotonicity at full strength is the intent margin flipping, and it is absent at half strength. **(D)** A ceiling on chain length improves safety and welfare together.

also delivers the same unsafe frequency: the gap is below 0.01 for five of the seven ceilings and at most 0.018, with a mean of 0.005. The agreement holds across variations in ϕ, ε, the organisational benefit and the ceiling, where the mean gap ranges from 0.001 to 0.016. The floor has no effect at all while it lies under $\phi^{\bar{D}} = 0.178$, the attribution a chain already reaches at the ceiling, and begins to bite as soon as it binds, at $a_{\min} = 0.22$; $a_{\min} = 0.31$ halves the harm and $a_{\min} = 0.5$ recovers 86% of the welfare that separates the baseline from strict liability.

This matters because the two instruments make very different demands on a regulator. A depth cap has to be enforced against a quantity that is hard to observe from outside and easy to restructure, namely how many hands an instruction passed through. A floor is a statement about liability alone: it says that no chain, however long, reduces the principal's share below $a_{\min}$, and it needs no one to count the layers. Where the floor cannot follow is the shallow end. Even $a_{\min} = 1$ leaves mean depth at 1.99, because with full attribution the erosion cost by itself still permits two hand-offs, so the floor spans the frontier down to a ceiling of two and no further.

6.6. Robustness

The 2×2 was recomputed across 43 parameter cells: the prize and the private setback risk of the race, both readings of what a setback destroys, the organisational benefit and the coordination cost of a taller chain, the population size, the selection intensity, the depth ceiling, the liability level, and the choice between the finite-population and replicator readings. The interaction is positive in 88% of them, with median +0.292 and interquartile range $[0.242, 0.297]$; the

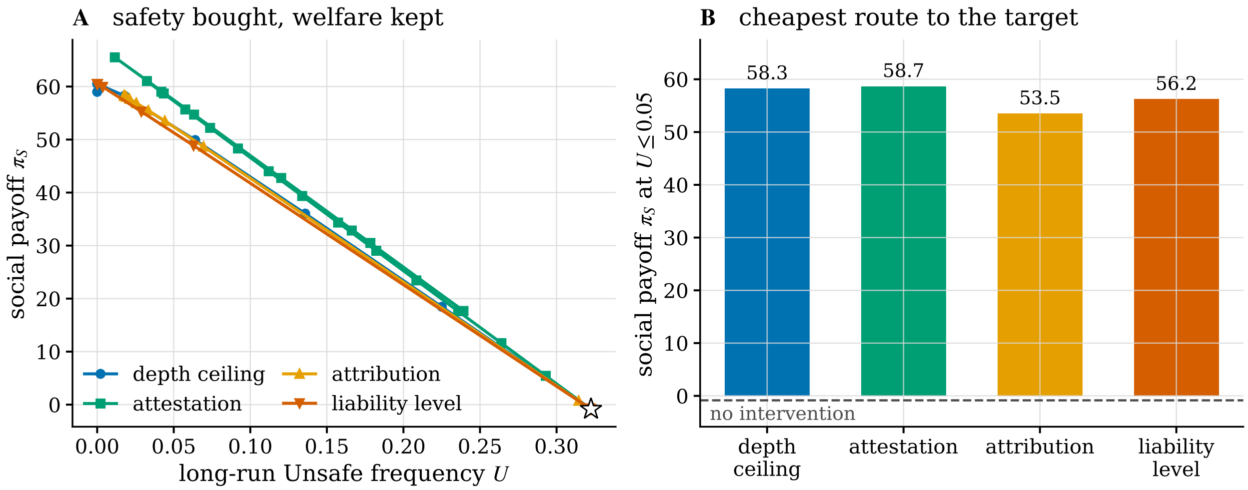

Figure 7: What each instrument costs. **(A)** The frontier each instrument traces in the plane of social payoff against long-run unsafe frequency, starting from the untreated baseline (star). **(B)** Social payoff at the weakest setting of each instrument that reaches an unsafe frequency of 0.05. Raising the liability level is the least efficient route, because depth has already discounted the margin it acts on.

Table 6

A floor under attribution reproduces a depth cap. For each ceiling $\bar{D}$, the attribution floor $a_{\min}$ delivering the same social payoff, and the Unsafe frequency each delivers. The two instruments trace the same frontier: the mean gap is 0.005 and the largest is 0.018, at the shallow end a floor cannot reach.

	depth cap		matched floor		
$\bar{D}$	U	π_S	$a_{\min}$	U	π_S
0	0.0000	59.0	1.00	0.0176	58.3
1	0.0000	60.5	1.00	0.0176	58.3
2	0.0178	58.3	0.87	0.0178	58.3
3	0.0640	49.9	0.49	0.0638	49.9
4	0.1359	36.0	0.35	0.1362	36.0
5	0.2251	18.5	0.28	0.2240	18.7
6	0.3225	-0.8	0.00	0.3225	-0.8

median interaction share is 0.91. The mean delegation depth is higher under per-layer attribution than under pass-through in every single cell. The negative cells are all in the prize-and-risk sweep and are ceiling effects, where the joint outcome is already at or near an unsafe frequency of one and cannot rise further.

The parameters of the evolutionary process deserve to be read cell by cell, since they are the ones a reader has least reason to accept on faith. The joint cell is 0.3225 at all three population sizes $Z \in \{50, 100, 200\}$ and moves only from 0.3166 to 0.3225 as the selection intensity varies by a factor of twenty over $\beta \in [0.01, 0.2]$; the interaction stays between +0.240 and +0.301. What the process parameters do move is the attribution-only cell, from 0.0622 at $\beta = 0.01$ down to 0.0040 at $Z = 200$, because weak selection and small populations let a mildly disadvantaged design persist. Since that cell is a component of the additive prediction, weak selection makes the interaction look smaller while leaving the joint outcome where it was.

The drift kernel is the one dimension along which the result genuinely changes, and the direction is instructive (Figure 8B). Replacing graded erosion by *collapse*, in which a lost specification is replaced by the unconditionally unsafe design rather than by the next rung down, all but eliminates the effect: the joint cell falls to 3×10^{-5} and mean depth to 0.0002. Delegation simply does not happen, because a chain that fails catastrophically is one whose principal

Table 7
Two kernel families at a fixed spectral gap. Both leave the second eigenvalue of the hand-off kernel at $1-\varepsilon$; the first removes the direction of an erosion event and the second enlarges it. Removing the direction leaves the cascade in place and eventually makes it worse, while enlarging the event destroys it at a threshold.

family	t	drift	$m(1)$	U	interaction	$\bar{d}$
direction	0.00	0.75	0.000	0.3225	+0.2934	6.00
direction	0.10	0.68	0.065	0.3209	+0.2984	6.00
direction	0.20	0.60	0.129	0.3204	+0.3073	6.00
direction	0.25	0.56	0.160	0.3205	+0.3087	6.00
direction	0.30	0.52	0.191	0.3207	+0.3091	6.00
direction	0.40	0.45	0.251	0.3215	+0.3100	6.00
direction	0.50	0.38	0.310	0.3227	+0.3112	6.00
direction	0.60	0.30	0.367	0.3242	+0.3127	6.00
direction	0.70	0.22	0.423	0.3270	+0.3155	6.00
direction	0.80	0.15	0.477	0.3325	+0.3210	6.00
direction	0.90	0.07	0.529	0.3867	+0.3752	6.00
direction	1.00	0.00	0.580	0.5170	+0.5055	6.00
severity	0.00	0.75	0.000	0.3225	+0.2934	6.00
severity	0.10	0.83	0.209	0.3955	+0.3839	6.00
severity	0.20	0.90	0.411	0.4601	+0.4486	6.00
severity	0.25	0.94	0.510	0.4249	+0.4133	5.30
severity	0.30	0.97	0.607	0.0219	+0.0104	0.27
severity	0.40	1.05	0.797	0.0058	-0.0057	0.06
severity	0.50	1.12	0.980	0.0019	-0.0097	0.02
severity	0.60	1.20	1.157	0.0006	-0.0109	0.00
severity	0.70	1.27	1.327	0.0002	-0.0113	0.00
severity	0.80	1.35	1.491	0.0001	-0.0114	0.00
severity	0.90	1.43	1.649	0.0001	-0.0115	0.00
severity	1.00	1.50	1.800	0.0000	-0.0115	0.00

will not build it. Replacing graded erosion by *uniform* noise, which has the same spectral gap but no direction, makes matters worse rather than better, with a joint cell of 0.517.

Three named kernels differ in more than one respect at once, so we separate the properties with two one-parameter families. The first runs from the ladder to the uniform kernel and takes the mean signed drift from 0.75 rungs per erosion event down to 0; the second runs from the ladder to collapse and takes it from 0.75 up to 1.5. Every drift kernel along both families has second eigenvalue zero, so both hold the second eigenvalue of $\boldsymbol{M}$ at exactly $1-\varepsilon$ and with it the asymptotic rate of forgetting.

The two families do not both hold the specification half-life of Proposition 2 fixed, and the difference is worth naming. That half-life is a property of the intact-arrival probability, which is $(1-\varepsilon)^d$ only while the drift kernel keeps a zero diagonal. The second family keeps one, so its half-life stays at 3.11 hand-offs. Along the first the diagonal fills in, the intact-arrival probability becomes $\frac{3}{4}(1-\varepsilon)^d + \frac{1}{4}$, which tends to $\frac{1}{4}$ rather than to zero because an eroded specification can be restored by chance, and the half-life rises to 4.92. Fidelity improves along the first family while the outcome gets worse, which is the section's point in miniature.

Along the first, the joint cell stays between 0.320 and 0.327 for four fifths of the range before rising to 0.517 at pure noise, and mean depth sits at the ceiling throughout. Removing the direction of the drift never helps and eventually hurts. Along the second family the picture is not gradual at all. The joint cell first rises to 0.460, and then, between mixture weights 0.25 and 0.30, mean depth falls from 5.30 to 0.27 and the joint cell to 0.022. Nothing about the spectral gap has changed, and the drift has become *more* directed rather than less.

Two conclusions follow, and neither is available from the three named kernels. The spectral gap is not the driver: it is identical in all twenty-four cells of Table 7 while the joint outcome ranges from 3×10^{-5} to 0.517. Directionality is not the driver either, and does not even order the outcomes, since collapse is the most directed kernel we consider and the safest of the three. What orders them is how much harm a *single* hand-off already causes. Under the ladder the first

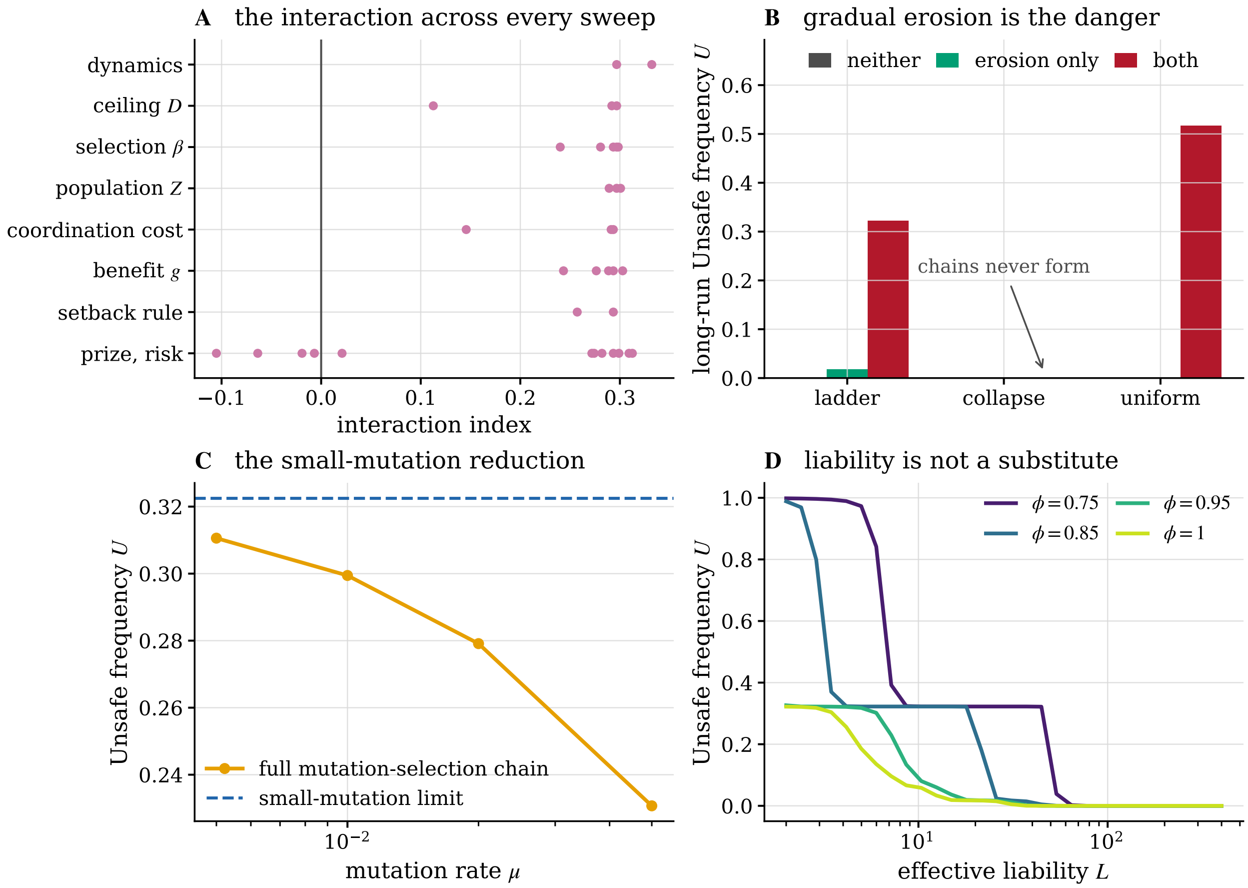

Figure 8: Robustness. **(A)** The interaction index across every sweep; each dot is one parameter cell. **(B)** Drift kernels. Graceful erosion produces the cascade; catastrophic erosion deters delegation altogether; undirected noise is worse than graded erosion, so the effect is not about the direction of the slippage. **(C)** The small-mutation limit against the full mutation-selection chain on a reduced design space. **(D)** Long-run unsafe frequency against the effective liability at four attribution retentions; a lower ϕ shifts the whole curve rather than tilting it.

hand-off is free, $m(1) = 0$, because one lost clause turns AS into CS, which is still safe against a safe opponent, and the chain is built because its first layer costs nothing. Along the severity family the collapse of depth happens exactly where $m(1)$ crosses the interval $[0.51, 0.61]$, and the direction family, whose $m(1)$ never exceeds 0.58, never collapses. The property that makes a delegation technology safe is neither fidelity nor unbiasedness: it is that the first thing to go wrong is expensive.

A last ablation asks whether a taller hierarchy being genuinely harder to run would close the shelter without any intervention. It does not, and the margin is wide. Raising the quadratic coordination cost c leaves the unsafe frequency at 0.322 and mean depth at the ceiling all the way to $c = 0.16$, and depth is still 5.95 at $c = 0.20$. It falls to 4.50 only at $c = 0.25$, which is exactly the point where $B(\bar{D}) = g\bar{D} - c\bar{D}^2$ reaches zero: cost prevents the shelter only once it has removed the entire organisational benefit of the deepest chain, not merely made the last layer unprofitable. The reason is that the organisational benefit is not the main private return to depth. Against a resident AS@$\bar{D}$ at $c = 0.14$, the race payoff of the focal chain rises monotonically with depth from 41.3 to 48.1, because an eroded chain executes more aggressive designs and those earn more in the race, while the attributed liability peaks at 11.4 at depth four and falls to 10.3 at the ceiling. Of the private gain the deepest chain enjoys, the organisational benefit supplies less than a third, and a cost instrument acts on that third alone.

7. Discussion

The model says something specific about where the governance of agentic systems is currently pointed. Almost all of the instruments now being built read or act at the two ends of a chain: evaluations and safety training at the model, and liability, disclosure and procurement rules at the deploying principal. Both ends are exactly where the model says the leverage is weakest. Regulation is written on the same plan, since the European Union's Artificial Intelligence Act allocates duties along a value chain by the role a party occupies and by the modification it makes, not by the number of hand-offs that separate an instruction from the action it eventually produces [57].

At the principal end, Theorem 4 says that a liability rule written in terms of what a principal is responsible for has a depth beyond which it stops binding, and Corollary 6 says where that depth is. Raising the penalty does not move the failure depth much, because it enters logarithmically; restoring attribution does, because it enters through $\log \phi$. The practical form of that observation is unwelcome for a regulator: partial pass-through is worth very little, and recovering 85% of responsibility at each of six hand-offs recovers 38% overall and, in our baseline, changes nothing at all. What the floor result adds is that an accountability mechanism does not have to achieve near-complete tracing to be worth deploying; it has to *guarantee a minimum*. The right figure of merit for such a mechanism is therefore not the average share of responsibility it traces but the smallest share it guarantees at the longest chain it permits. The legal analogue is familiar: where an injurer's assets fall short of the harm it can do, the remedy that works is not a larger penalty but a minimum the injurer is required to carry at all times [40, 58].

At the model end, the declaration result says that a safety commitment issued at the top of a long chain is close to uninformative about the ecosystem below it, and not because anyone is lying. In the baseline equilibrium the declaration is as strong as the language permits and the behaviour is 74% different from it. Organisation theory calls that outcome decoupling and treats the good faith with which a formal structure and a divergent practice are maintained side by side as the normal case rather than the exception [59]; what the model adds is a selection mechanism that produces it from delegation depth alone.

The non-monotonicity result deserves its own caution. Making hand-offs more faithful can raise realised harm, because the over-specification that erosion had forced becomes unnecessary and the population relaxes its declared intent by one clause, and a clause dropped from CS costs far more than a clause dropped from AS. The general form of the caution is familiar from second-best welfare economics, where satisfying one more of the conditions for an optimum while others remain out of reach is in general neither necessary nor sufficient for an improvement [60]. In this model only the instruments acting on the incentive to delegate are monotone.

Finally, the kernel comparison suggests a design principle that does not come from the liability literature at all. A specification that degrades gracefully is more dangerous than one that fails loudly, because graceful degradation lets a principal build a chain whose failures it never sees. Systems whose specifications fail closed, refusing rather than approximating when a clause is lost, restore the private cost of depth that per-layer attribution removes.

What would have to be measured. We have no empirical estimate of ε or ϕ for any deployed system, and the numbers here should not be read as if we had. The two are not the same kind of quantity. Both ε and the drift kernel are properties of a transmission technology and are directly observable in an instrumented pipeline: fix a task, write the principal's specification as an explicit inventory of clauses, log the instruction each agent passes to the next, and label each logged instruction for which clauses it still carries. The empirical hand-off matrix $\hat{M}$ is then the transition matrix of that labelled sequence between consecutive layers, and ε and $\hat{D}$ are read off it directly. Section 6.6 says which feature of $\hat{D}$ matters, and it is neither the spectral gap nor the direction: the quantity to estimate is the harm a single hand-off already causes, which needs only the same task run at depth zero and depth one with the unsafe actions counted in both. By contrast ϕ is not a property of the system at all but of the liability regime around it, so it has to be estimated from practice rather than from logs, as the share of remediation cost actually borne by the commissioning principal in incidents at each chain depth. The model asks less of these estimates than it might appear, because its claims are ordinal and the policy-relevant questions are which side of the severity threshold a transmission technology sits on and whether attribution is bounded below.

7.1. Limitations

The interaction layer is a stylised two-player race with four reduced strategies, inherited deliberately and without modification from Fernández Domingos and Han [16]. It is not a model of any deployed system, and the absolute numbers should be read as the behaviour of that game rather than of a market.

The erosion kernel is exogenous. We describe how a population behaves under a given transmission technology and not how the technology itself is selected; a model in which providers compete on fidelity would let ε co-evolve with d, and the non-monotonicity of Section 6.4 suggests that the outcome would not be a simple race to the top. Our chains are also linear, whereas real agentic systems branch, and a branching chain aggregates several eroded specifications rather than one.

Geometric attribution is a modelling choice, and Section 6.5 reports what happens when it is replaced. What the model requires is only that attribution vanish in the depth, and an equal split of the blame across the chain delivers the same equilibrium as ϕ^d even though it decays polynomially. Real attribution is nonetheless lumpier than any of the rules we try: a single identifiable intermediary may absorb everything, or pass-through may hold at some depths and not at others, and we have not modelled attribution that depends on which layer failed rather than on how many there were.

On the computational side, the small-mutation limit is exact only as $\mu \to 0$ and Table 3 quantifies the residual at finite μ on a reduced design space; we have not established a bound on that residual for the full space, where the full chain cannot be formed to compare against. The revision clock is Poisson, and the Erlang alternative of Kara et al. [52] is the natural next test. Finally, the harm scale h enters the social functional and therefore the welfare comparisons, and we do not know it; we have kept the primary outcome, the long-run unsafe frequency, free of welfare weights for that reason.

8. Conclusion

Delegation depth is not neutral plumbing between a principal and an action. It erodes the specification geometrically and it dilutes the attributed harm geometrically, and because the second effect is unbounded while the first saturates, depth is a liability shelter that no level of liability closes. What makes the combination dangerous is not either mechanism but the loss of a response: a principal that is charged for what its chain does answers unreliable hand-offs by keeping the chain short, and per-layer attribution is precisely what stops it. In our baseline that lost response accounts for 91% of the damage, and it produces an equilibrium in which every principal issues the safest instruction available and the ecosystem executes something 74% different.

Placing a structural variable under selection alongside strategy is what makes these statements available, and it is also what makes the process expensive to compute, because the design space is a product. The scheme of Section 5 keeps the whole calculation exact given the payoff matrices: the interaction tensor is summed over the horizon law rather than sampled, the mutation–selection chain on 3.0×10^{27} population states is replaced by an embedded chain on 28 designs, and the fixation probabilities that chain needs are evaluated in $O(Z)$ operations from a sum whose terms are shifted before exponentiation. The same three ingredients transfer unchanged to any evolutionary game whose strategy space is a product of a behavioural coordinate with a structural one.

The instruments that follow from the model are not the ones currently receiving attention. Limiting how many hand-offs may separate an instruction from an action, and verifying specifications at the agent that acts rather than at the party that spoke, are the two levers that behave predictably here. Raising penalties works least of all, because depth has already discounted the margin they act on. Restoring attribution hand-off by hand-off works only when it is nearly complete, but that is a fact about the geometric rule rather than about attribution: what the shelter needs is for the principal's share to keep falling, and a rule that stops falling closes it. A floor under attribution buys what a cap on chain length buys and asks less of whoever has to enforce it.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All results are deterministic given the seed used for the replicator start measure. The code, the generated tables, the numerical benchmarks of Section 5 and the figure pipeline are openly available under the MIT licence at `https://github.com/trungkiet2005/delegation-cascade`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used a large language model assistant in order to draft and copy-edit prose and to refactor analysis scripts. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No text, figure or numerical result was accepted without verification against the model and the code released with the paper.

CRediT authorship contribution statement

Trung-Kiet Huynh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Dao Sy Duy Minh:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Chi-Nguyen Tran:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

A. The stationary regime in closed form

The derivation of (12) is standard and is repeated here so that the constants in Algorithm 1 can be read off. In a population of Z individuals holding k copies of the invader I and $Z-k$ of the resident R, an update picks an ordered pair of distinct individuals uniformly, so the probability that an R individual is picked to revise and an I individual to imitate is $\frac{Z-k}{Z}\cdot\frac{k}{Z-1}$, and symmetrically in the other direction. With the Fermi rule the two transition probabilities are

$$T^{+}(k)=\frac{k(Z-k)}{Z(Z-1)}\,\frac{1}{1+e^{-\beta(\pi_I(k)-\pi_R(k))}},\qquad T^{-}(k)=\frac{k(Z-k)}{Z(Z-1)}\,\frac{1}{1+e^{\beta(\pi_I(k)-\pi_R(k))}},\tag{15}$$

with π_I and π_R as in (11). Their ratio is $T^{-}(k)/T^{+}(k)=e^{-\beta(\pi_I(k)-\pi_R(k))}$, in which the common sampling factor cancels, and the classical absorption formula for a one-dimensional birth-death chain with absorbing boundaries at $k=0$ and $k=Z$,

$$\rho_{R\to I}=\left[1+\sum_{m=1}^{Z-1}\prod_{k=1}^{m}\frac{T^{-}(k)}{T^{+}(k)}\right]^{-1},$$

gives (12) directly. Because $\pi_I-\pi_R$ is affine in k, the exponent $\theta_m=-\beta\sum_{k\le m}(\pi_I(k)-\pi_R(k))$ is a quadratic polynomial in m and the whole vector $(\theta_1,\dots,\theta_{Z-1})$ is one cumulative sum, which is the $O(Z)$ claim of Section 5.3. Two limits are useful as unit tests of an implementation: at $\beta=0$ every ratio is one and $\rho=1/Z$, and for a neutral game the same holds at every β.

Under the small-mutation limit the embedded chain (10) is irreducible whenever every $\rho_{i\to j}$ is strictly positive, which holds at finite β for every pair, so its stationary distribution is unique. Section 5.3 is the reason that positivity has to be preserved numerically rather than merely asserted: a routine that returns exactly zero for a strongly disadvantaged mutant makes the embedded chain reducible in floating point, and the stationary vector of a reducible chain is no longer unique.

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